

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			glucose + oxygen $\longrightarrow$ carbon dioxide + water (+ energy / ATP) 1 mark for the reactants 1 mark for the products	2			2		
	(b)	(i)	I II	Both correct for 1 mark 3.5 - 4.5 minutes 14.5 -15.5 minutes			1	1	1	
		(ii)		As heart rate increases breathing rate increases / as one increases the other increased / higher the heart rate, higher the breathing rate / ORA (1) Any one (x1) from Breathing rate increases after heart rate increases / ORA (1) Breathing rate decreases after heart rate decreases / ORA (1) Changes in breathing rate lag behind changes in heart rate / owtte / ORA (1) Breathing rate takes longer to return to {normal / resting} / breathing rate does not return to normal but heart rate does / ORA (1)		2		2		
		(iii)	I	119 = 2 marks If incorrect award 1 mark 140 x 85 / 100 140 x 0.85 85% of 140		2		2	2	2

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			II	Horizontal line drawn at 119 (length of line is not important) Tolerance less than one small square 116- 123 Ecf			1	1		1
		(iv)		Lactic acid {build up / produced} / anaerobic respiration creates lactic acid (1) Ref to oxygen debt (1) ref to oxygen required to {break down / get rid of / remove} lactic acid / oxygen is required to convert lactic acid to carbon dioxide and water (1)	2			2		
	(c)			Because less ATP produced / ORA) / only produces 2 ATP / Anaerobic produces 2 ATP but aerobic produces {more / 38} incomplete breakdown of glucose during anaerobic respiration / less energy <u>released</u> / ORA	1			1		
				Question 7 total	5	4	2	11	3	3